

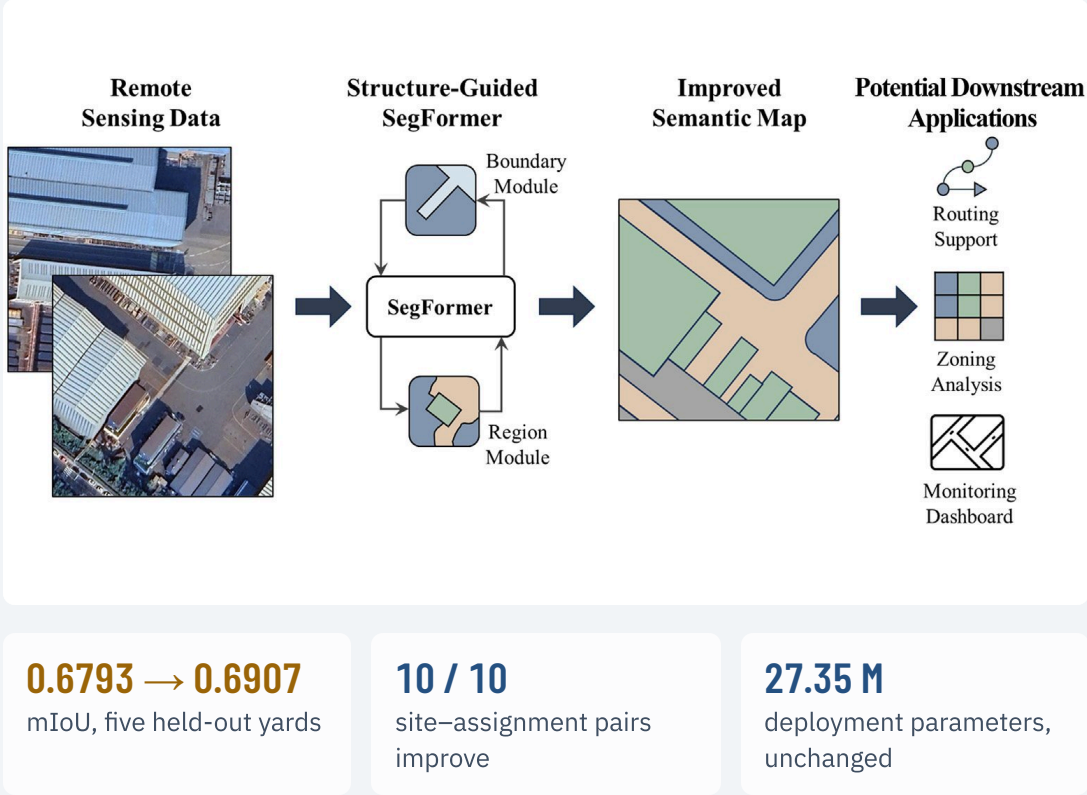
Mapping roads, yards and buildings in shipyards the model has never seen

Structure-guided cross-site segmentation:
boundary and region constraints used in
training only, deployment model unchanged

*Structure-guided cross-site semantic segmentation of
shipyard remote-sensing imagery*

Chunrui Na, Qing Wang, Jingzheng Yao*, Youngrong Kim*
Harbin Engineering University · Nanyang Technological University · HEU Yantai Research
Institute
Ocean Engineering 368 (2026) 128269 · doi:10.1016/j.oceaneng.2026.128269

Figure: paper Fig. 1 (imagery Google, Airbus). * Corresponding authors.



The paper in one minute

1

The task

Split shipyard imagery into background, road, yard open area and production building. In another yard, layout and appearance change: class transitions blur and large regions fragment.

2

The method

Two training constraints on SegFormer-B2: a boundary constraint derived from the label morphology, and region consistency through eight learnable prototypes. Both are removed at inference.

3

The results

Five held-out yards × three seeds: mIoU 0.6793 → **0.6907**, boundary F1 0.7190 → 0.7393, fragmentation 0.0775 → 0.0631. All 10 site-assignment comparisons improve.

4

The cost

The deployed model is the baseline: 27.35 M parameters, 92.1 GFLOPs. Only training takes slightly longer: 32.20 → 34.18 s per epoch.

Boundaries fix the transitions, regions fix the interiors: complementary, and free at deployment.

The order of this briefing (click to jump)

1

Background and difficulty

What a shipyard semantic map is for; why a new yard is hard

p. 4

2

Data and protocol

Eight yards, 988 patches; five held out in turn

p. 6

3

Method

Boundary and region-consistency constraints

p. 8

4

Main results

Overlap, classes, boundaries and fragments, per site, longer training

p. 11

5

Mechanism and engineering value

Prototypes learn; free deployment; road connectivity

p. 16

6

Qualitative results

Five yards compared; local gains and residual errors

p. 19

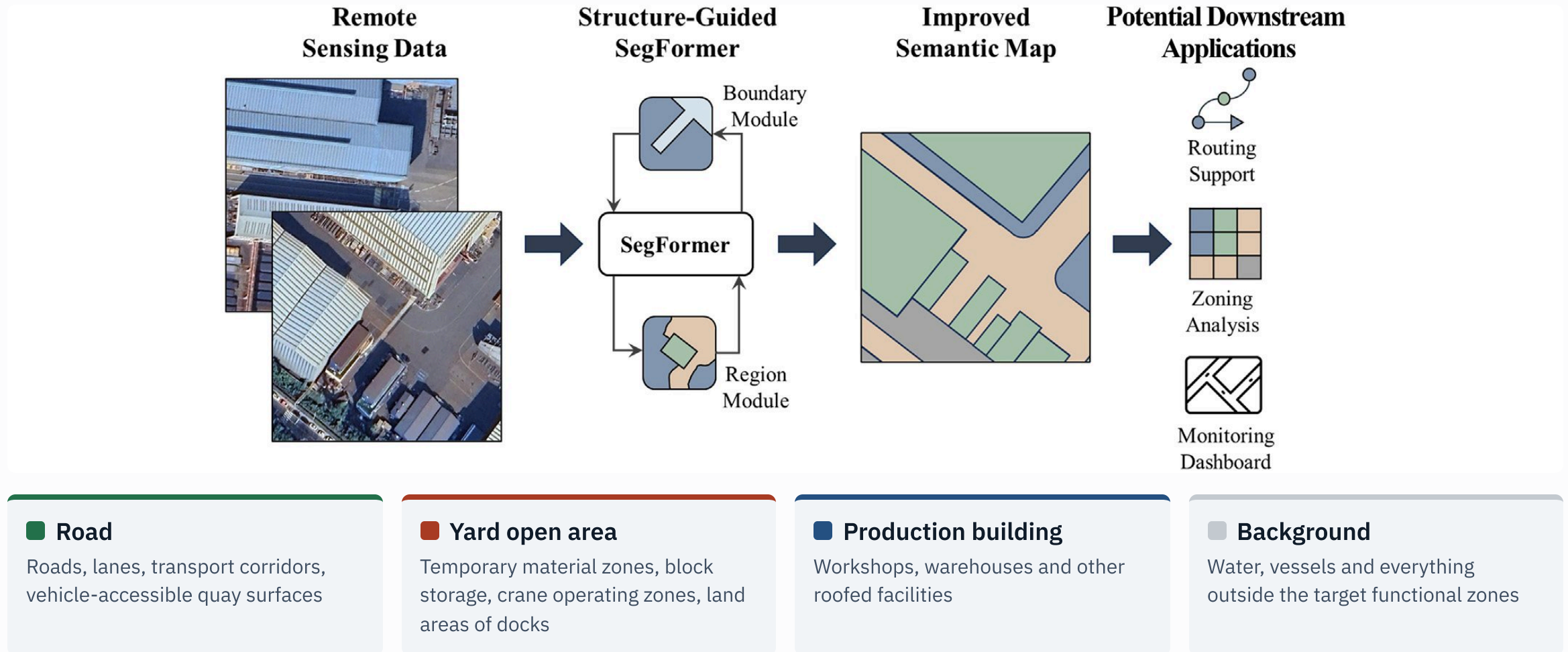
7

Conclusions

Contributions and place in the thesis

p. 21

A shipyard semantic map turns imagery into a spatial layer for digital twins and transport planning



Sources: paper Fig. 1 (imagery Google, Airbus) and the class definitions of §3.2. The map can feed route-network extraction, zoning, facility localisation and spatial indexing; those downstream modules are outside this paper.

The difficulty: in another yard, layout, facilities, roads and appearance all change

1

Scene-level distribution shift

Building density, road geometry, yard texture and organisation differ from yard to yard

2

Weak boundaries between functional regions

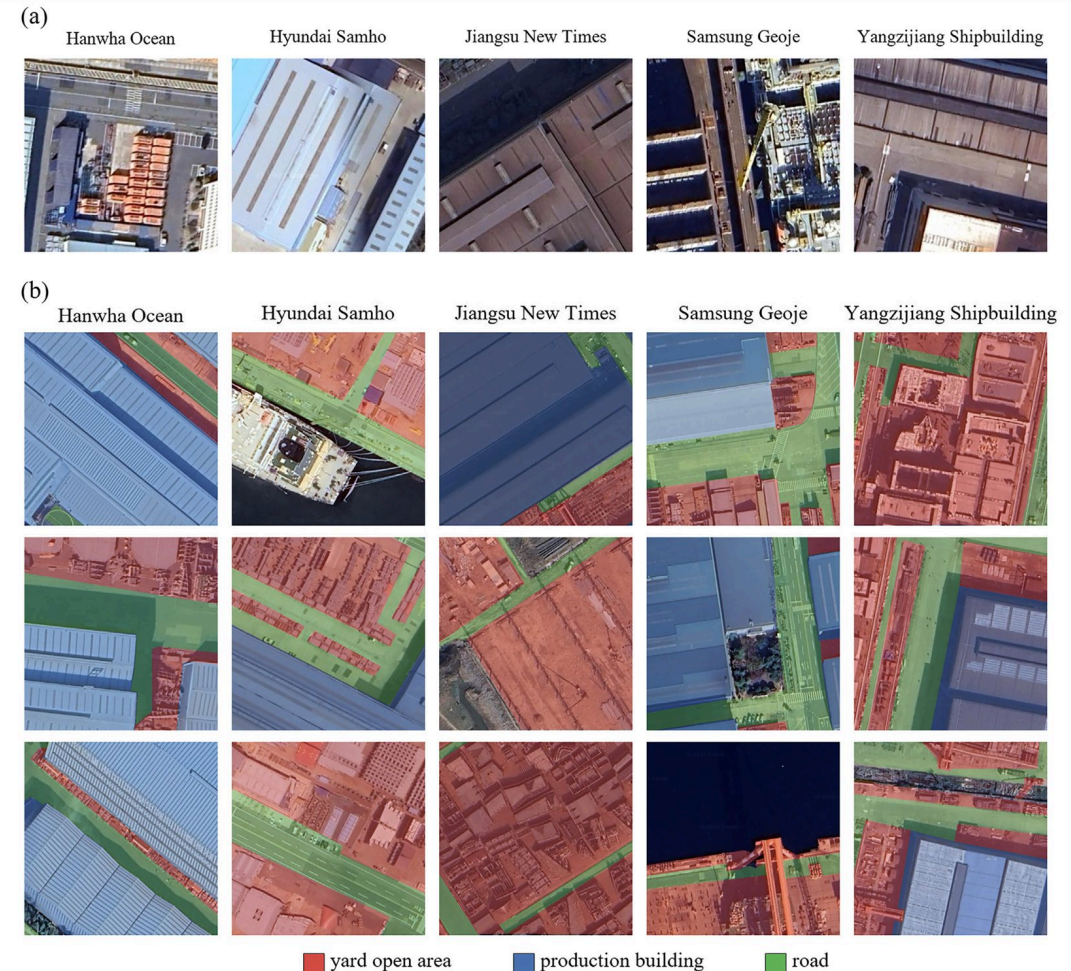
Road–yard and building edges are where errors concentrate

3

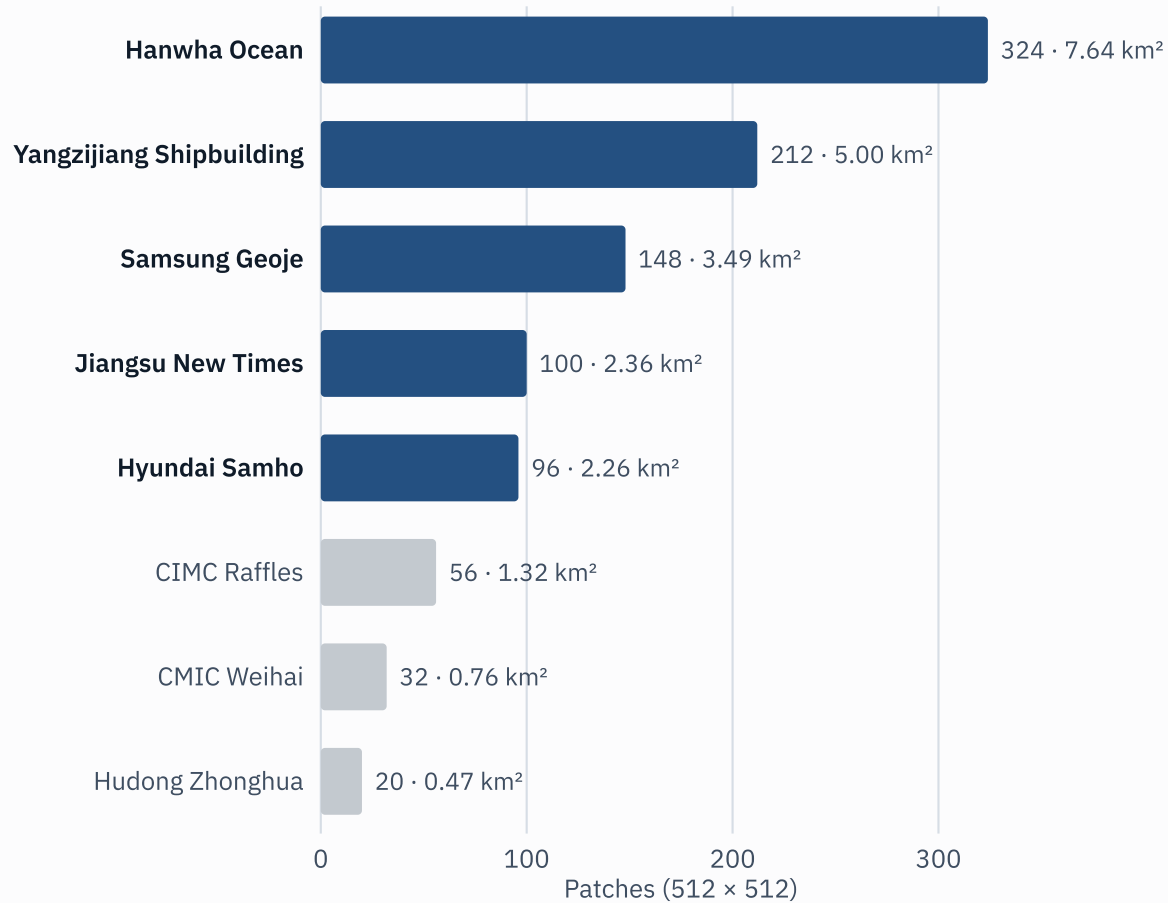
Fragmented interiors

Uniform yards and roofs are predicted as scattered small pieces

**No image of the target yard is used for training or adaptation:
this is domain generalisation, not adaptation.**



Dataset: eight yards, 988 patches, about 23.3 km², four roughly balanced classes



■ Held-out test yard (in turn) ■ Training only

Pixel share of the four classes



247 → 988

1024 × 1024 source images cut into four 512 × 512 quadrants; about 0.30 m/pixel; exported from Google Earth Pro in 2025

0.921 mIoU · 0.887 BF1

Three independent annotators with review and adjudication; pairwise agreement on a 99-patch QC subset (10.02%)

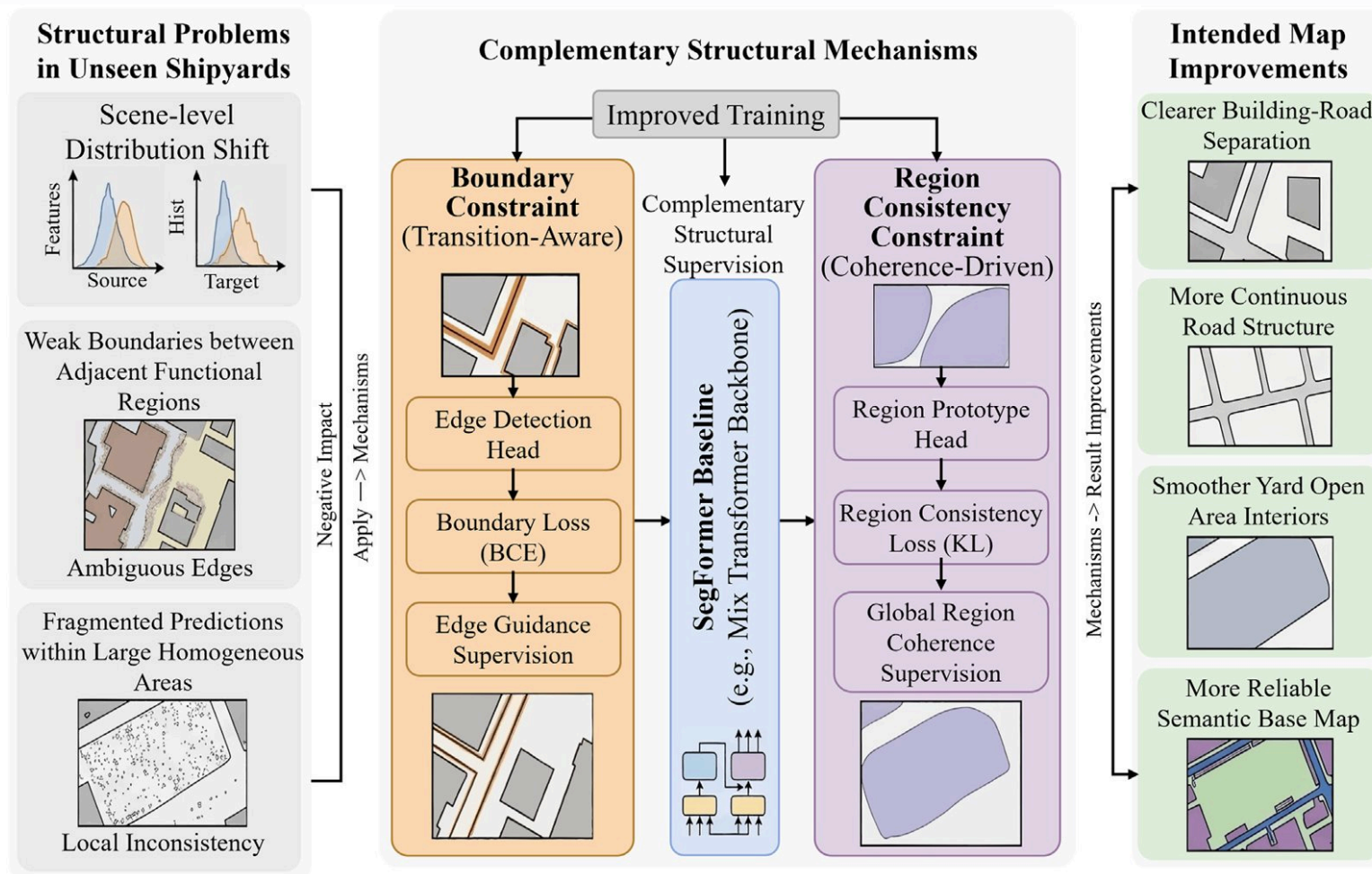
Protocol: five yards held out in turn, with two validation-yard assignments

TEST YARD	VALIDATION YARD	TRAIN / VAL / TEST PATCHES
Hanwha Ocean	Samsung Geoje	516 / 148 / 324
Hyundai Samho	Hanwha Ocean	568 / 324 / 96
Jiangsu New Times	Hyundai Samho	792 / 96 / 100
Samsung Geoje	Yangzijiang Shipbuilding	628 / 212 / 148
Yangzijiang Shipbuilding	Jiangsu New Times	676 / 100 / 212

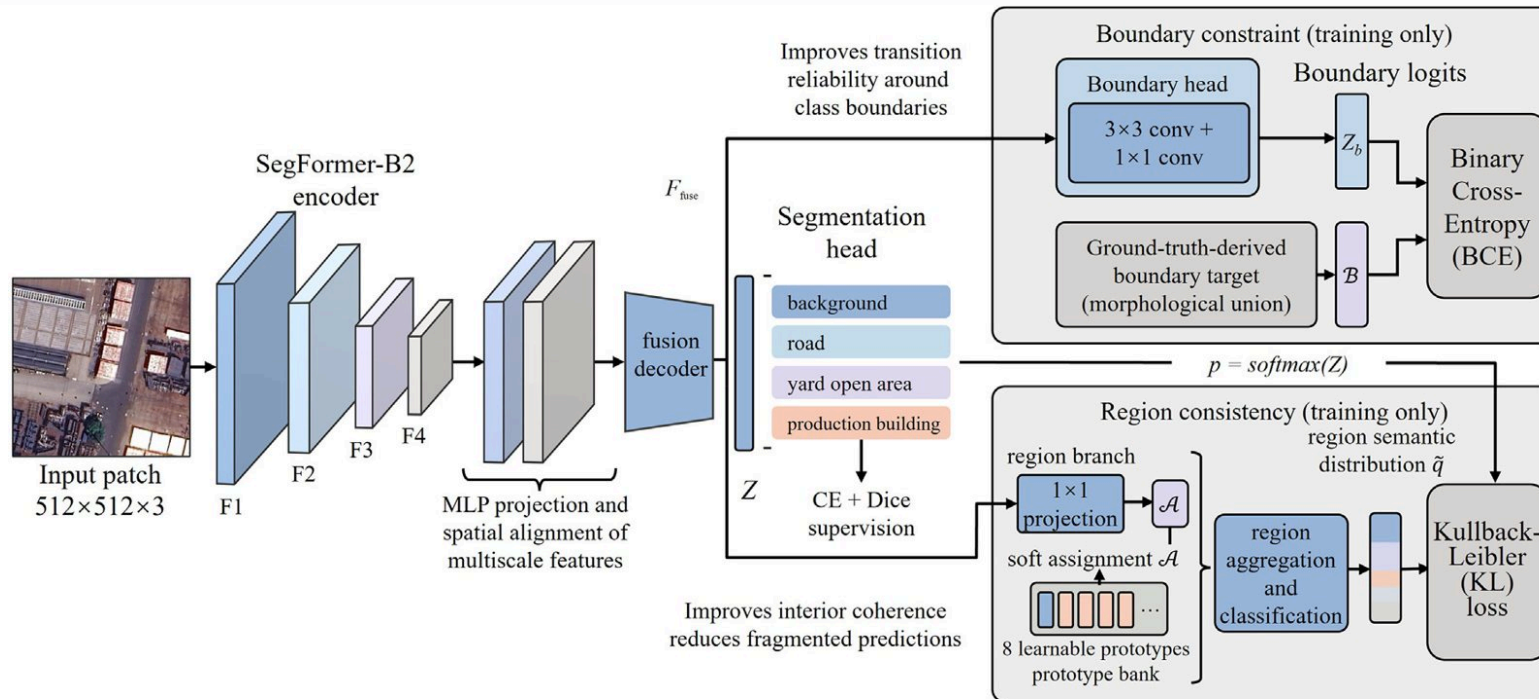
The table shows assignment B, used for the main comparison: one test yard, one validation yard and six training yards per fold.

- **Leave one shipyard out:** each of five well-sampled yards is the test yard in turn; no target-yard image is used for training or adaptation
- **Two validation assignments, A and B:** fixed before training, each a five-cycle; every test yard has a different validation partner in A and B
- **Three seeds** (3407, 5519, 7823) → 15 runs per method and assignment; checkpoints chosen by validation mIoU
- **Two budgets:** a controlled 30-epoch comparison (same-backbone ablation) and a 120-epoch external comparison

The idea: a boundary constraint for transitions, a region-consistency constraint for interiors



Architecture: both auxiliary branches exist only in training and are removed for inference



Joint training objective

$$\mathcal{L} = \mathcal{L}_{ce} + \lambda_{dice} \mathcal{L}_{dice} + \lambda_b \mathcal{L}_{boundary} + \lambda_r \mathcal{L}_{region}$$

$$\mathcal{L} = \mathcal{L}_{ce} + \lambda_{dice} \mathcal{L}_{dice} + \lambda_b \mathcal{L}_{boundary} + \lambda_r \mathcal{L}_{region} \quad \cdot \quad \lambda_{dice} = 1.0, \lambda_b = 0.5, \lambda_r = 0.1$$

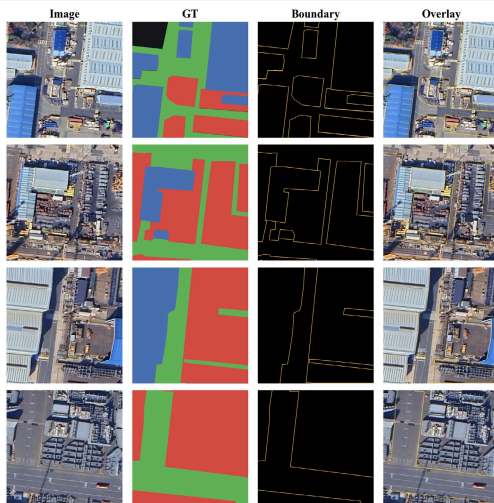
Three branches share the fused decoder features F_{fuse} : segmentation head (CE + Dice), boundary head (BCE), region-consistency branch (KL). The deployment graph remains SegFormer-B2.

How the two constraints are built: no extra labels, no change at inference

Boundary constraint: transitions

$$B = \max_c \text{clip}(\text{Dilate}(M_c) - \text{Erode}(M_c), 0, 1) \quad (3 \times 3)$$

- Boundary targets come straight from the label's morphological gradient, with no dedicated boundary labels; an ideal straight transition gives a band of about two pixels (0.6 m)
- A light boundary head (3×3 conv $256 \rightarrow 128$, BN, ReLU, 1×1 conv) under BCE



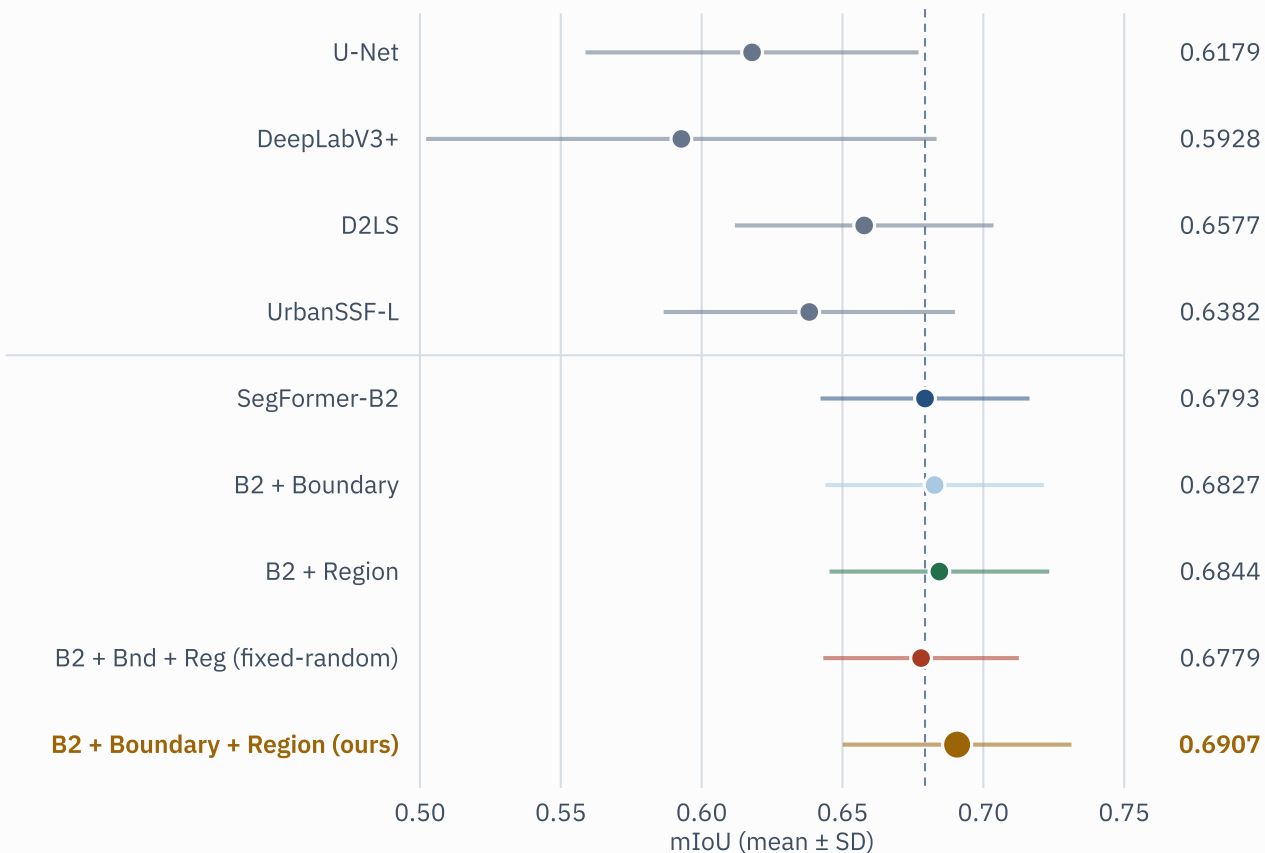
Region consistency: interiors

$$\tilde{q}_i = \sum_r a_{i,r} \text{softmax}(z_r) \quad \cdot \quad \mathcal{L}_{\text{region}} = \text{KL}(p_i \parallel \tilde{q}_i)$$

- **Eight learnable region prototypes** (8×256); cosine similarity of pixel features to prototypes (temperature $\tau = 0.07$) gives soft assignments a
- Region features aggregated by assignment $\rightarrow$ region classifier $\rightarrow$ projected back to each pixel as the region-level distribution $\tilde{q}$
- Pixel predictions p are pulled towards $\tilde{q}$ (KL), updating both sides, so predictions within a region stop fragmenting
- **Control:** prototypes sampled and then frozen (fixed-random), isolating prototype trainability

Main comparison: the joint model leads on all three overlap metrics; each branch alone also helps

mIoU fg_mIoU mDice



+0.0114

mIoU gain over SegFormer-B2 (+1.68%)

+0.0034 / +0.0051

mIoU gain of boundary-only / region-only: each helps, the joint model most

0.6898

First under assignment A too: mIoU 0.6777 → 0.6898

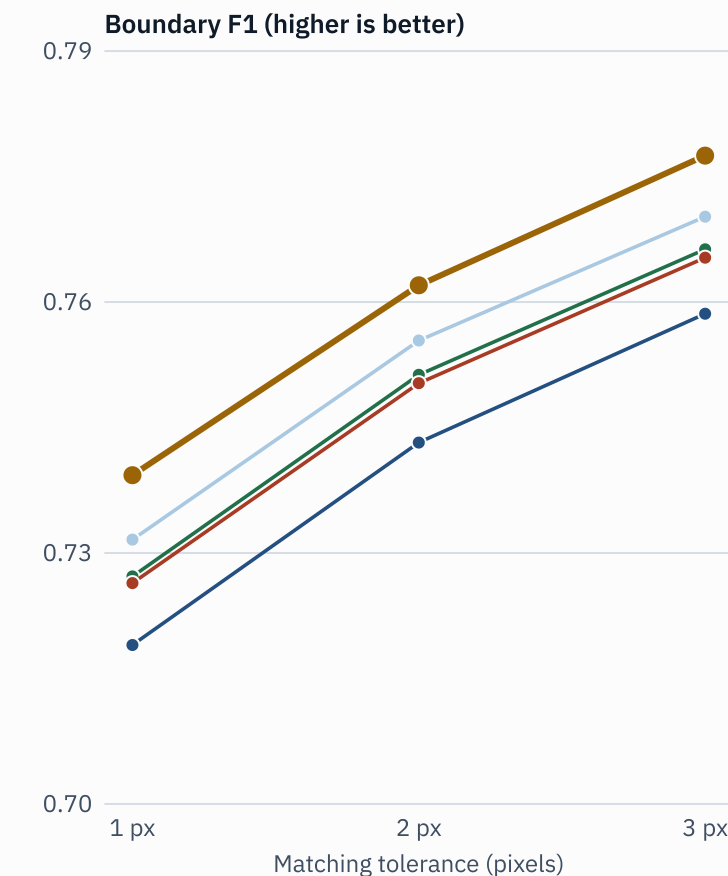
Highest IoU in all four classes, with the largest margins for buildings and roads

METHOD	BACKGROUND	BUILDING	ROAD	YARD	FG_MIOU	MIOU
U-Net	0.6242	0.7269	0.5516	0.5689	0.6158	0.6179
DeepLabV3+	0.6000	0.6885	0.5318	0.5509	0.5904	0.5928
D2LS	0.6514	0.7852	0.5854	0.6088	0.6598	0.6577
UrbanSSF-L	0.6358	0.7802	0.5562	0.5806	0.6390	0.6382
SegFormer-B2	0.6882	0.8020	0.6201	0.6069	0.6763	0.6793
B2 + Boundary	0.6938	0.8108	0.6225	0.6037	0.6790	0.6827
B2 + Region	0.6955	0.8105	0.6248	0.6068	0.6807	0.6844
B2 + Bnd + Reg (fixed-random)	0.6833	0.8141	0.6042	0.6100	0.6761	0.6779
B2 + Boundary + Region (ours)	0.6960	0.8237	0.6323	0.6108	0.6889	0.6907

Margins over the next-best method: building +0.0096, road +0.0075, yard +0.0008, background +0.0005. Darker cells are higher within their column.

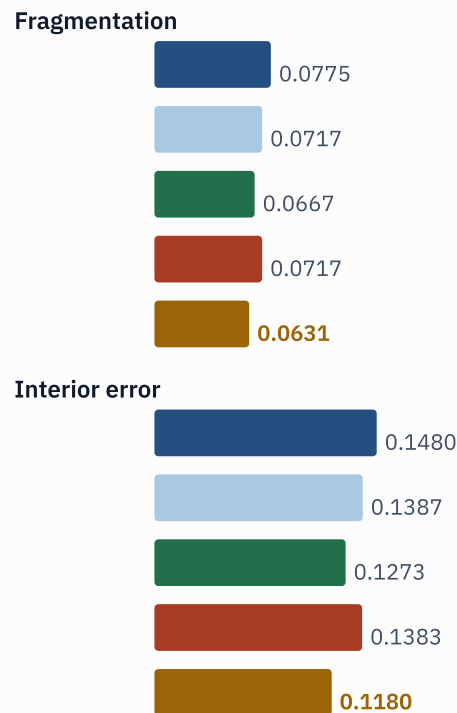
Source: paper Table 7 (class-wise and aggregate IoU over the 15 assignment-B runs, nine methods).

Structural quality: sharper boundaries, fewer fragments, fewer interior errors; the two branches complement each other



■ SegFormer-B2
 ■ B2 + Boundary
 ■ B2 + Region
 ■ B2 + Bnd + Reg (fixed-random)
 ■ **B2 + Boundary + Region (ours)**

Fragmentation and interior error (lower is better)



0.7190 → 0.7393

Boundary F1 at 1-pixel tolerance; also highest at 2 and 3 pixels

-18.6%

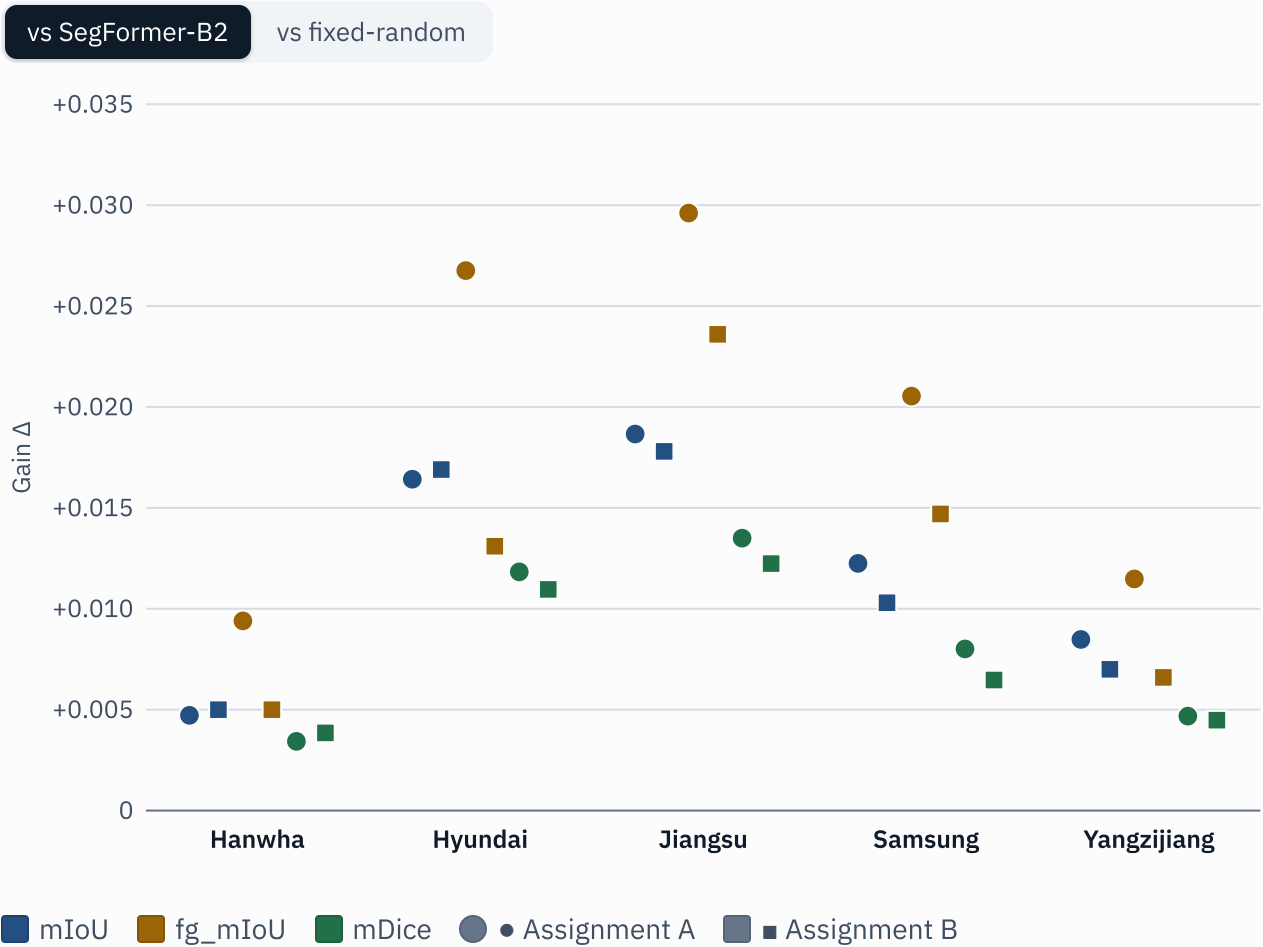
Small-component ratio (fragmentation) 0.0775 → 0.0631

-20.3%

Interior error rate 0.1480 → 0.1180; yard 0.158 → 0.116

Boundary-only helps boundary F1 more; region-only helps fragmentation and interior errors more; the joint model gets both.

Site by site: gains are positive for every held-out yard under both validation assignments



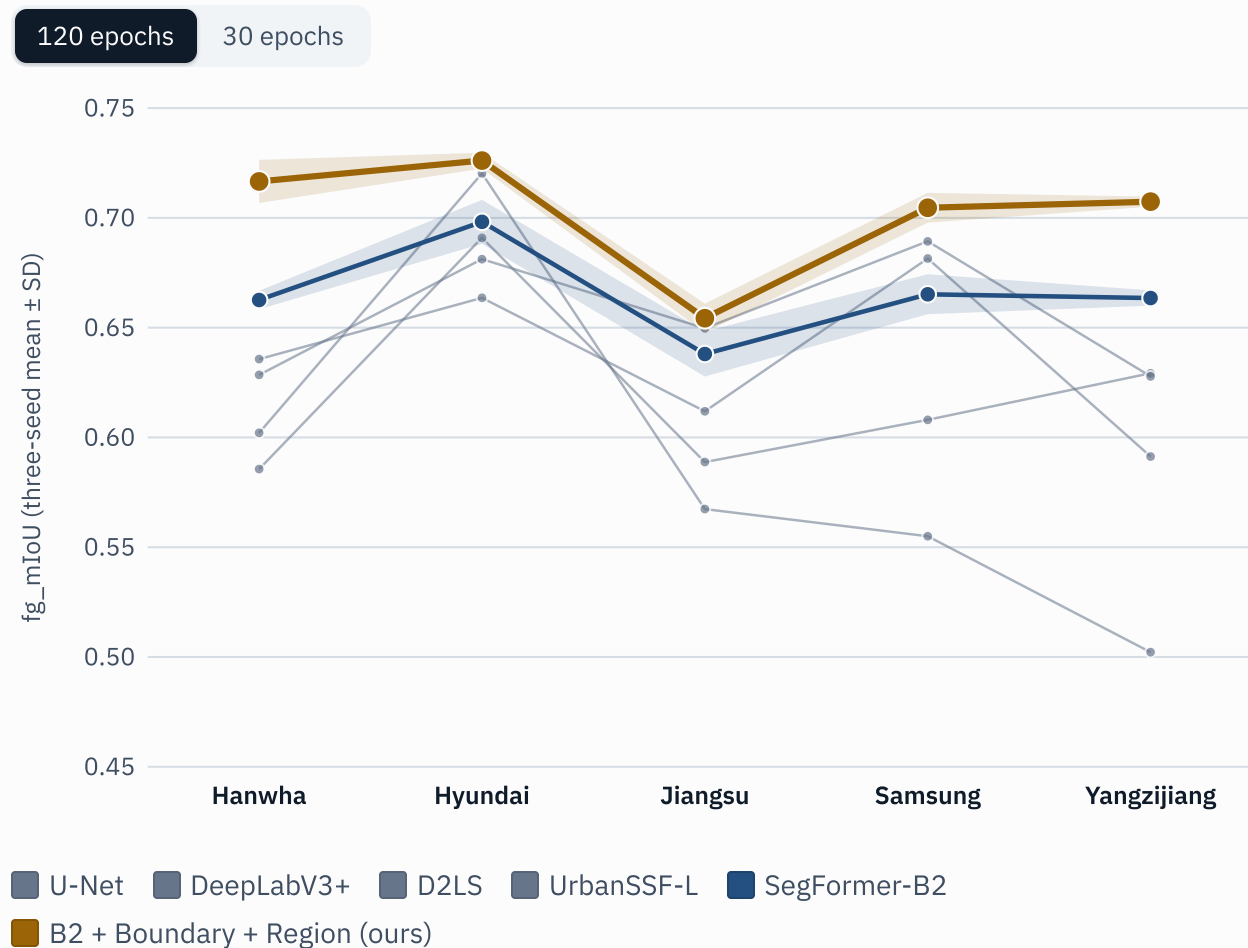
10 / 10

site-assignment comparisons, all positive on all three metrics

	SITE-LEVEL MEAN Δ	95% INTERVAL	POSITIVE YARDS	EFFECT SIZE D_z
mIoU	+0.01175	[0.00730, 0.01621]	5/5	2.06
fg_mIoU	+0.01608	[0.01002, 0.02214]	5/5	2.01
mDice	+0.00795	[0.00474, 0.01115]	5/5	1.95

Sources: paper Tables 10–12. Each point is one test yard's three-seed mean gain under one assignment. Site-level pairing averages both assignments and three seeds per yard, giving five paired differences; the 95% interval is a percentile interval over all 3,125 resamples. With $n = 5$ all positive, the exact two-sided Wilcoxon $p = 0.0625$, the smallest value five pairs can reach.

Longer training: at 120 epochs the advantage holds, above the baseline on all five yards



+0.0163 – +0.0540

fg_mIoU gain over SegFormer-B2 on the five yards: +0.0163 to +0.0540

0.6942

120 epochs: mIoU 0.6782 → 0.6942

0.7018

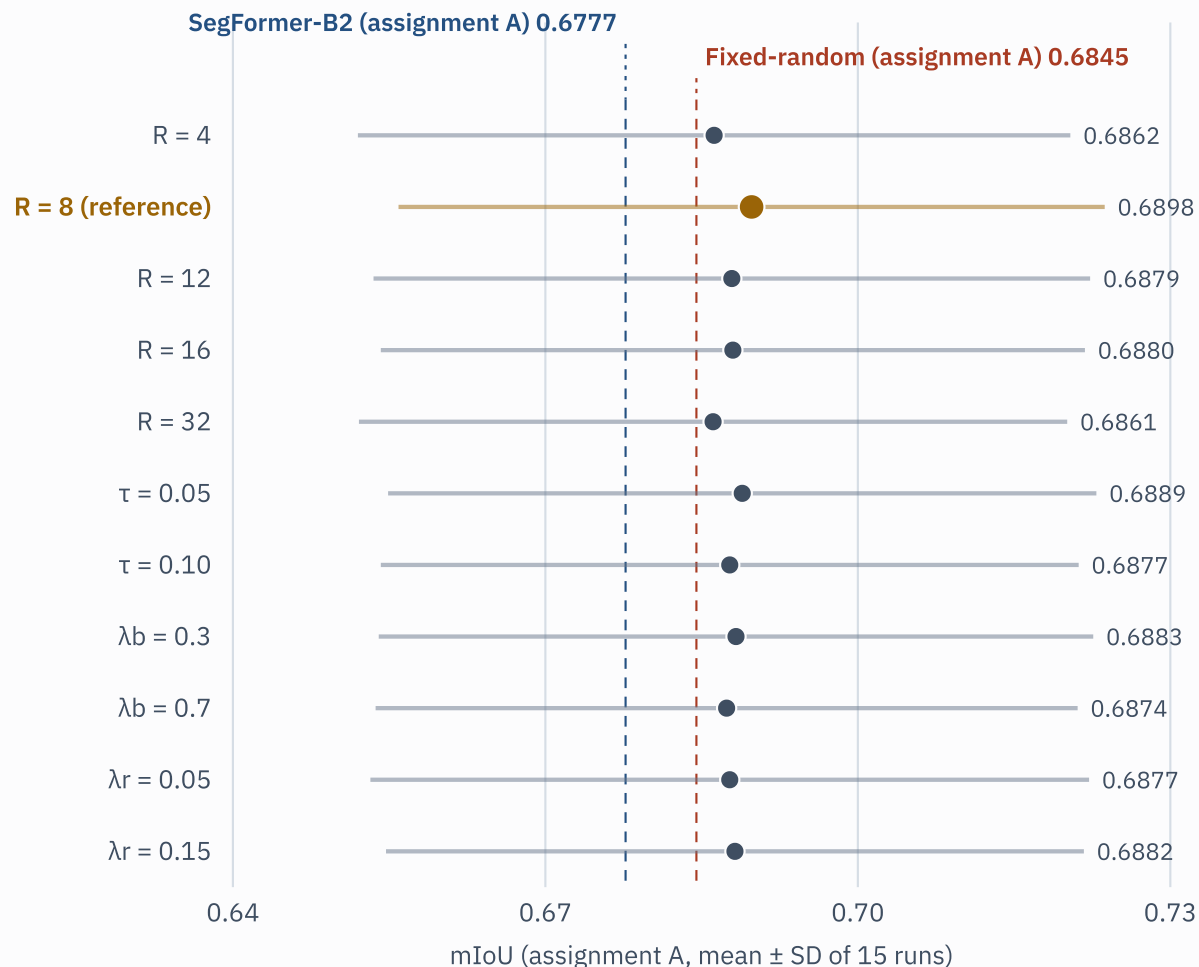
120 epochs: fg_mIoU 0.6655 → 0.7018

0.8103

120 epochs: mDice 0.8021 → 0.8103

Sources: paper Tables 15, 17, 18 and §5.7. At 120 epochs each method uses its own reference learning rate and a cosine schedule; across 0.5×, 1× and 2× learning rates the joint model ranks first in all three metrics at each multiplier, and its lowest mean over the grid exceeds every comparator's highest.

The prototypes do learn: learnable beats fixed-random, and nearby settings stay robust



+0.0091 mIoU

Site-level mean gain over fixed-random prototypes, 95% interval [0.0062, 0.0113], positive on 5/5 yards

0.512 → 0.802

Per-position maximum assignment: learned prototypes assign more decisively (normalised entropy 0.641 → 0.312)

7.56–7.95 / 8

Effective number of prototypes: all eight are used overall, no collapse

All 11 one-factor settings on the left beat both SegFormer-B2 and the fixed-random control.

Free at deployment: the inference model is exactly SegFormer-B2

27.35 M

deployment parameters (same as baseline)

92.1 G

FLOPs (same as baseline)

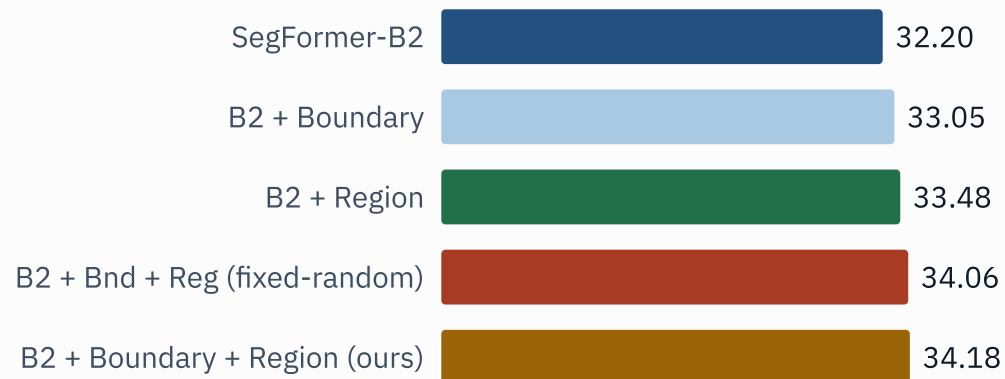
31.10 ms

latency per image; baseline 31.14 ms

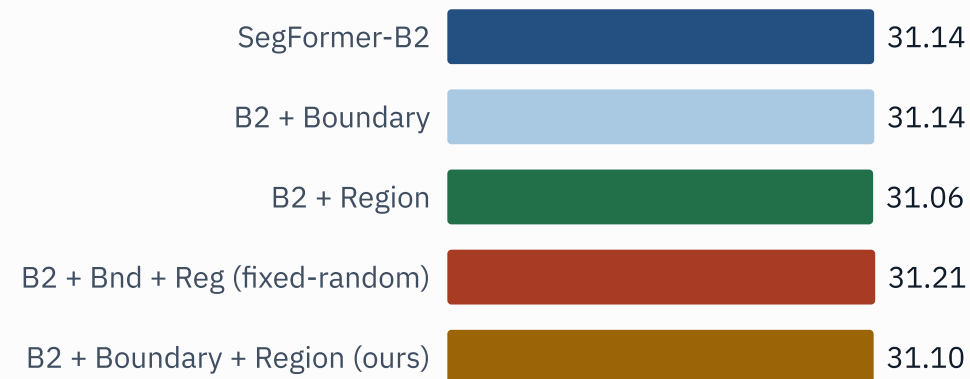
+6.1%

training per epoch 32.20 → 34.18 s;
364,165 auxiliary parameters exist only in training

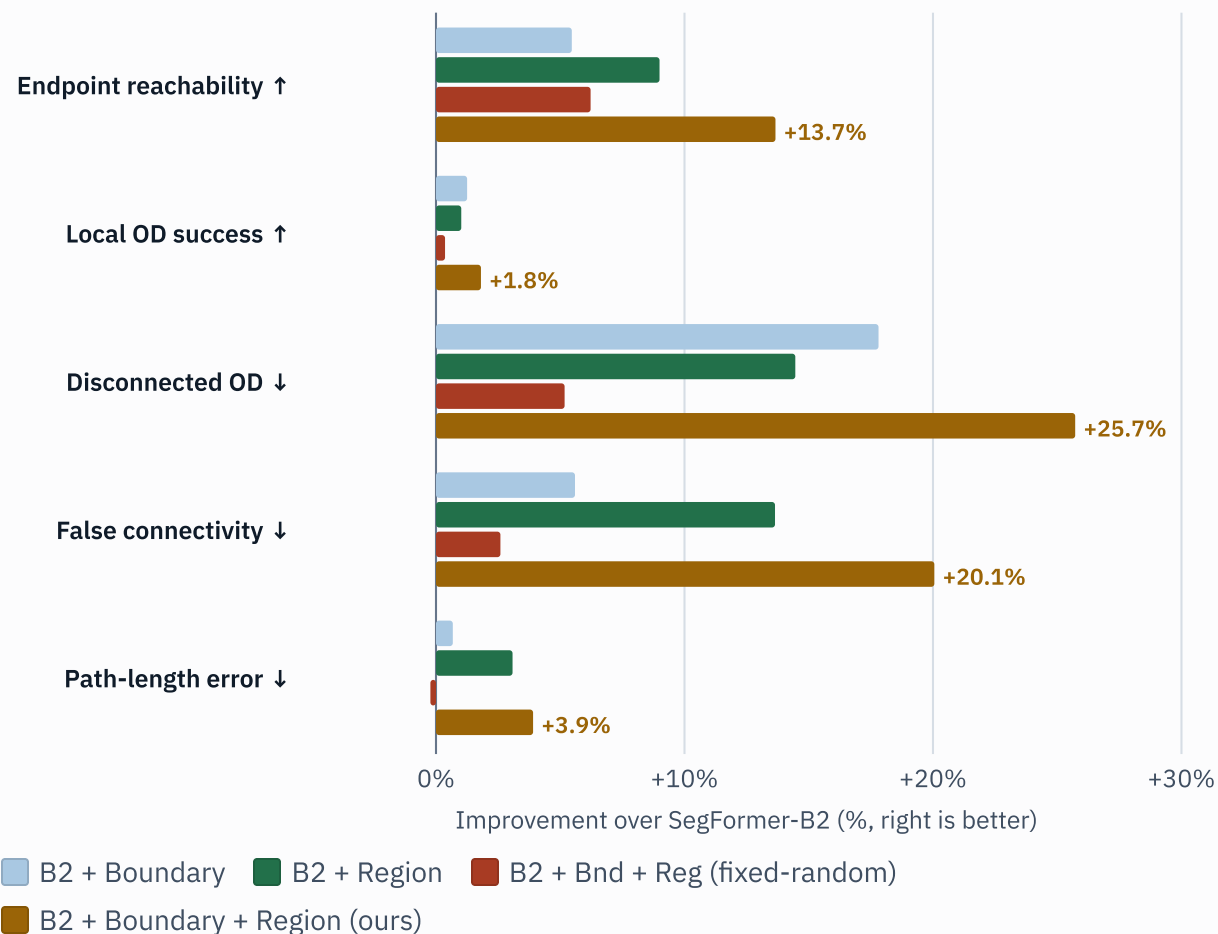
Training (s / epoch)



Inference latency (ms / image)



Towards road networks: better local connectivity, while long-range reachability still needs topology repair



6.57% → 4.88%

Local path failures (origin–destination pairs 4.8–19.2 m apart within one connected road)

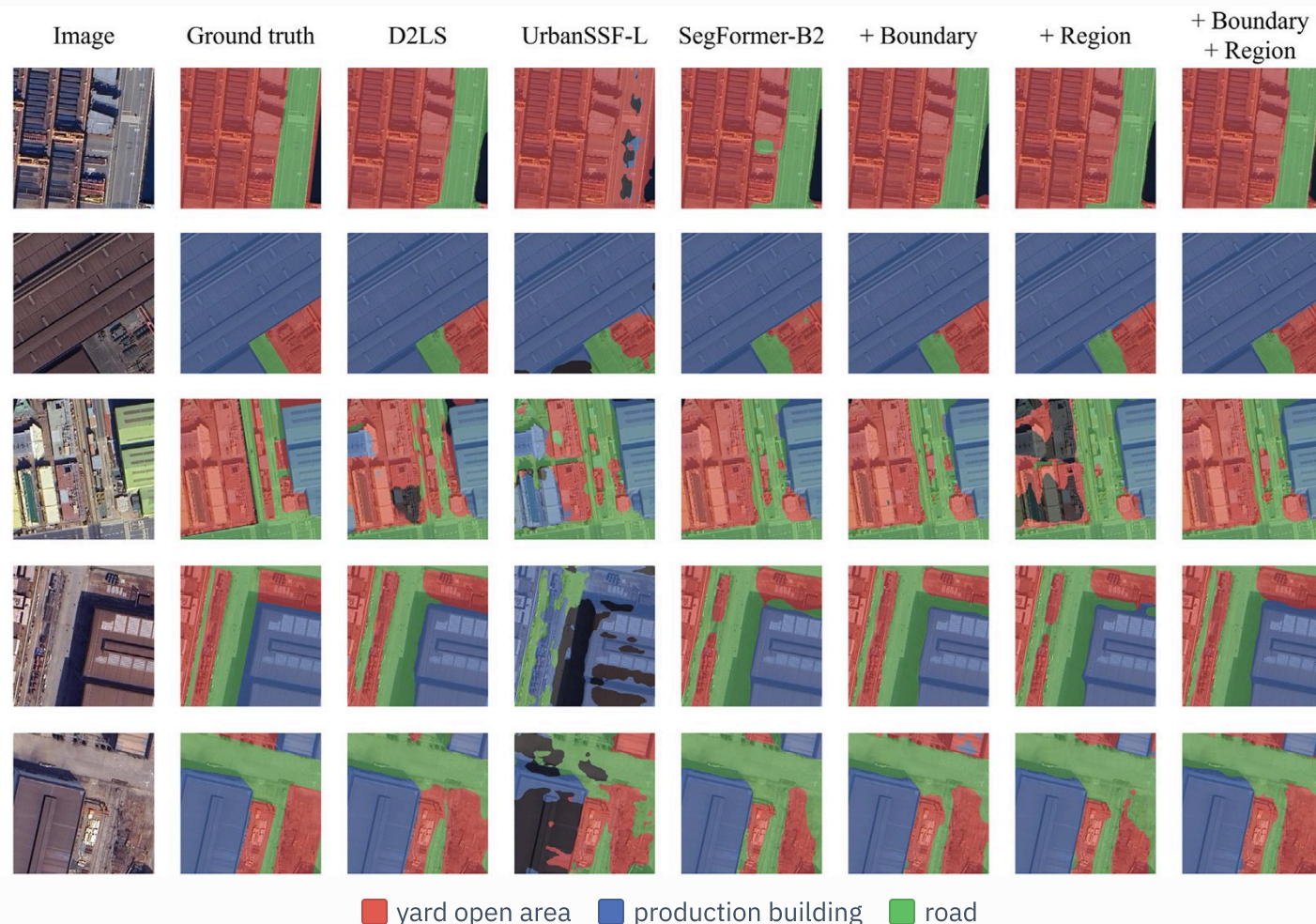
0.0733 → 0.0586

False connectivity: separate roads wrongly joined

0.2704

Long-range endpoint reachability (≥ 38.4 m): still low, so transport planning needs topology repair and GIS vectorisation first

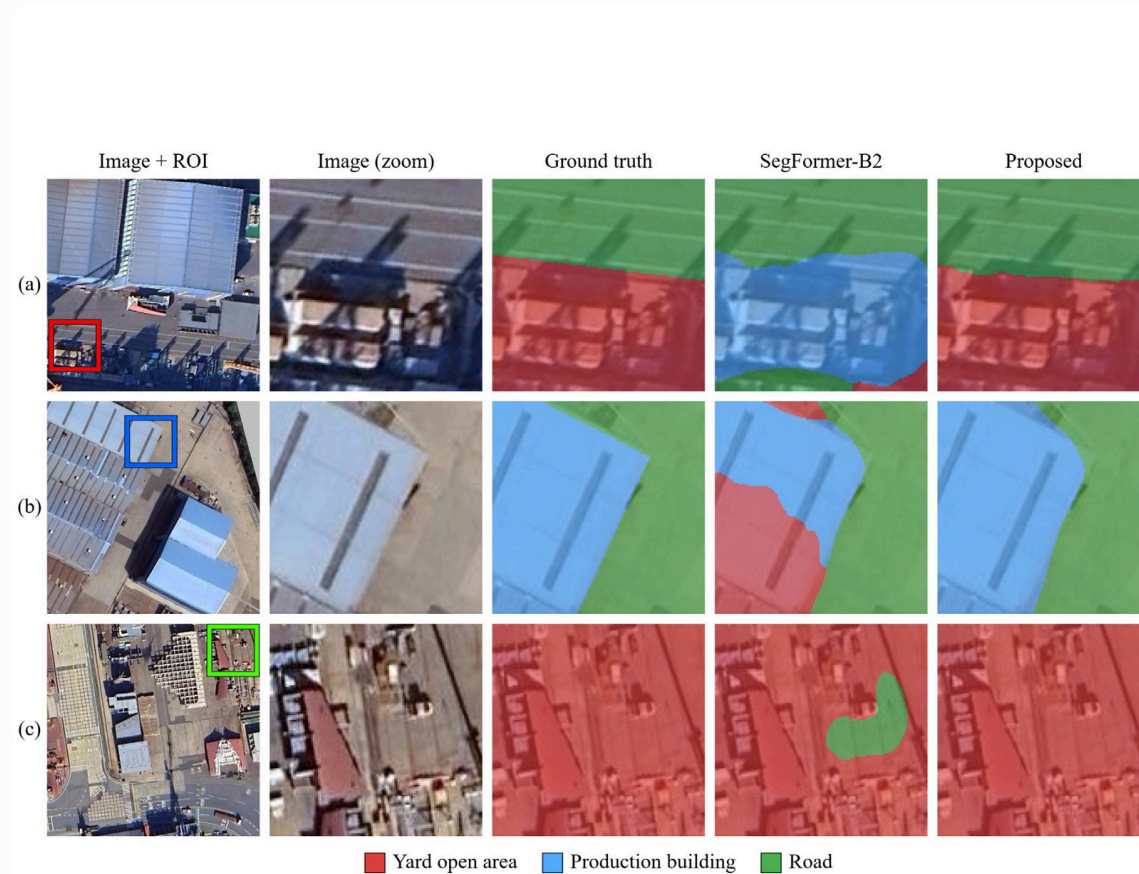
Qualitative comparison on the five held-out yards: clearer transitions, fewer interior fragments



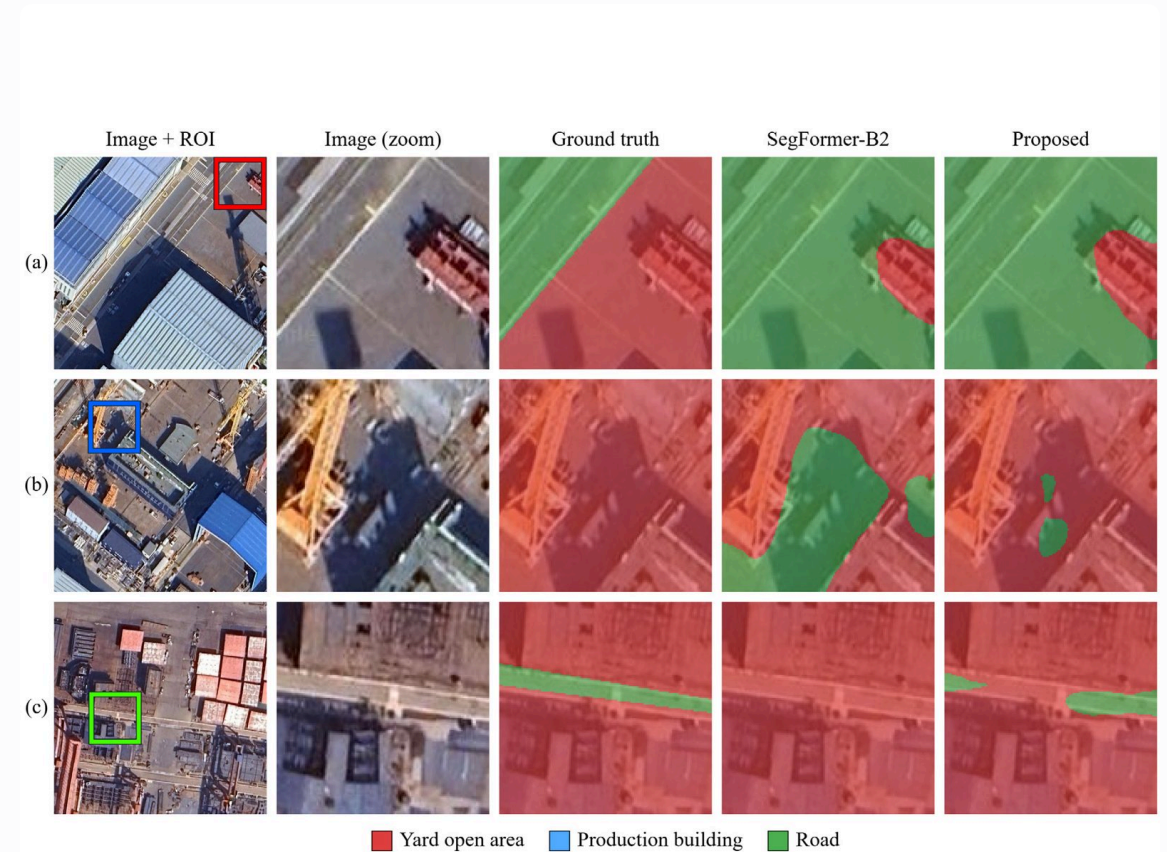
Source: paper Fig. 10 (30-epoch models; imagery Google, Airbus). Columns: image, ground truth, D2LS, UrbanSSF-L, SegFormer-B2, +Boundary, +Region, +Boundary+Region (ours).

Local improvements and the errors that remain

Gains: (a) road–yard interface (b) building boundary (c) yard interior



Residual errors: (a) road–yard confusion (b) shadow or occlusion (c) thin connection broken



Conclusions: complementary training constraints make cross-yard maps more accurate and cleaner at no deployment cost

1**A cross-yard protocol**

Five held-out yards, two validation assignments, three seeds

2**Two training constraints**

Morphological boundaries + prototype region consistency, removed at inference

3**Controlled ablations**

Separating boundary, region and prototype trainability

4**Complementary structural gains**

Interface and interior errors both fall; deployment graph unchanged

Its place in the thesis

It supplies semantic base maps that can speed up layout extraction for chapter 6's many yards; road topology is still traced as centrelines, with segmentation as support and citation.

Next

External tests across regions, sensors and seasons; whether topology-corrected maps improve route planning.

[Thesis briefing →](#)[Code →](#)

Boundaries fix the transitions, regions fix the interiors: complementary, and free at deployment.

Thank you.

References (1/2)

1. Na, C., Wang, Q., Yao, J., Kim, Y. (2026). Structure-guided cross-site semantic segmentation of shipyard remote-sensing imagery. *Ocean Engineering* 368, 128269. doi:10.1016/j.oceaneng.2026.128269
2. Bertasius, G., Shi, J., Torresani, L. (2016). Semantic segmentation with boundary neural fields. *Proc. CVPR*, 3602–3610.
3. Borse, S., Wang, Y., Zhang, Y., Porikli, F. (2021). InverseForm: a loss function for structured boundary-aware segmentation. *Proc. CVPR*, 5901–5911.
4. Chen, L.C., Zhu, Y., Papandreou, G., Schroff, F., Adam, H. (2018). Encoder-decoder with atrous separable convolution for semantic image segmentation. *Proc. ECCV*, 833–851. doi:10.1007/978-3-030-01234-2_49
5. Efron, B., Tibshirani, R. (1986). Bootstrap methods for standard errors, confidence intervals, and other measures of statistical accuracy. *Statistical Science* 1(1), 54–75. doi:10.1214/ss/1177013815
6. Han, Y.S., Lee, K., Nam, B., Kim, Y.O., Kim, Y., Yeo, H.B. (2024). Assessment of ship-block stacking status based on images obtained from drone and GIS data. *Int. J. Naval Architecture and Ocean Engineering* 16, 100583. doi:10.1016/j.ijnaoe.2024.100583
7. Hill, M.O. (1973). Diversity and evenness: a unifying notation and its consequences. *Ecology* 54(2), 427–432. doi:10.2307/1934352
8. Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian J. Statistics* 6(2), 65–70.
9. Milletari, F., Navab, N., Ahmadi, S.-A. (2016). V-Net: fully convolutional neural networks for volumetric medical image segmentation. *Proc. 3DV*, 565–571. doi:10.1109/3dv.2016.79
10. Qin, W., Song, Y., Zhu, H., Yu, X., Tu, Y. (2023). A novel shipyard production state monitoring method based on satellite remote sensing images. *Remote Sensing* 15(20), 4958. doi:10.3390/rs15204958
11. Ronneberger, O., Fischer, P., Brox, T. (2015). U-Net: convolutional networks for biomedical image segmentation. *Proc. MICCAI*, 234–241. doi:10.1007/978-3-319-24574-4_28

References (2/2)

12. Wang, C., Wang, K., Tao, J., Zhou, Y. (2020). Research on real-time optimal path planning model and algorithm for ship block transportation in shipyard. *J. Marine Science and Engineering* 8(12), 991. doi:10.3390/jmse8120991
13. Wang, J., Zheng, Z., Ma, A., Lu, X., Zhong, Y. (2021). LoveDA: a remote sensing land-cover dataset for domain adaptive semantic segmentation. *NeurIPS Datasets and Benchmarks*.
14. Wang, N., Yin, J., Khan, R.U. (2024). Risk coupling assessment of vehicle scheduling for shipyard in a complicated road environment. *J. Marine Science and Engineering* 12(4), 685. doi:10.3390/jmse12040685
15. Wang, Z., Yi, J., Chen, A., Chen, L., Lin, H., Xu, K. (2025). Accurate semantic segmentation of very high-resolution remote sensing images considering feature state sequences (UrbanSSF). *ISPRS J. Photogrammetry and Remote Sensing* 220, 824–840. doi:10.1016/j.isprsjprs.2025.01.017
16. Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin* 1(6), 80–83. doi:10.2307/3001968
17. Xie, E., Wang, W., Yu, Z., Anandkumar, A., Alvarez, J.M., Luo, P. (2021). SegFormer: simple and efficient design for semantic segmentation with transformers. *NeurIPS* 34, 12077–12090.
18. Yuan, Y., Chen, X., Wang, J. (2020). Object-contextual representations for semantic segmentation. *Proc. ECCV*, 173–190. doi:10.1007/978-3-030-58539-6_11
19. Zhang, T.Y., Suen, C.Y. (1984). A fast parallel algorithm for thinning digital patterns. *Communications of the ACM* 27(3), 236–239. doi:10.1145/357994.358023
20. Zou, X., Li, Y., Zhang, S., Li, K., Wang, S., Tao, P., Xing, J., Lang, C. (2025). Dynamic dictionary learning for remote sensing image segmentation (D2LS). *Proc. ICCV*, 22457–22466.